Bode Part 2: Sketching and Reading Bode Plots

EGR 345

GVSU

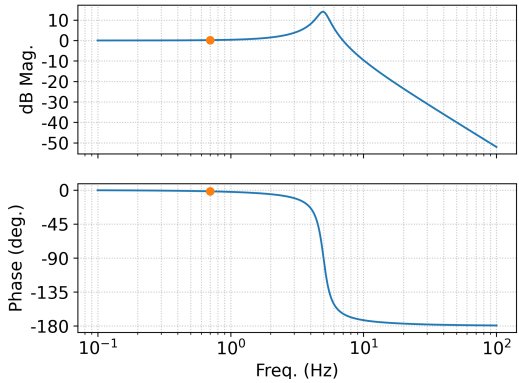
Outline

- quick review: frequency response from Bode
- Bode sketching and reading

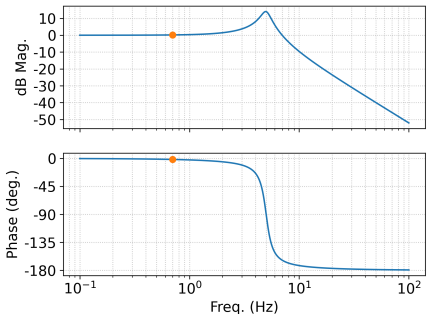
Quick Review

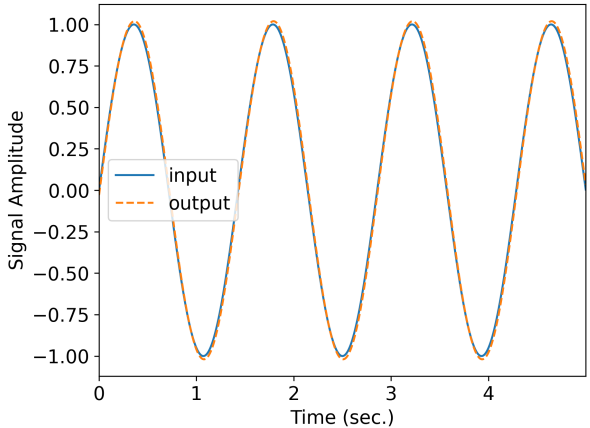
Fixed Sine vs. Bode 1

- what would you expect the steady-state input/output relationship to look like?



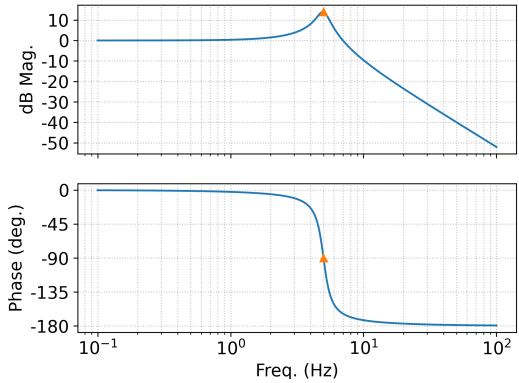
Fixed Sine vs. Bode 1 (answer)



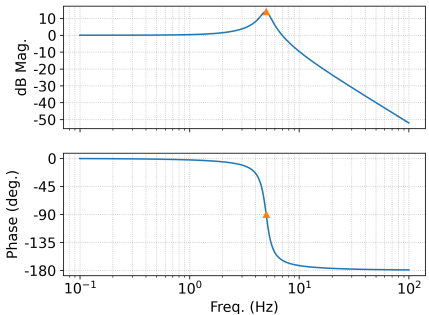


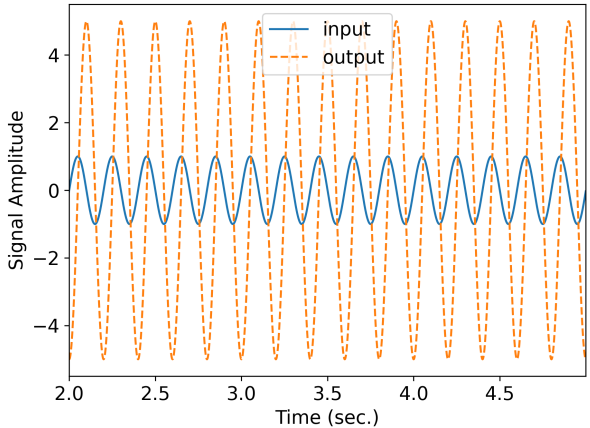
Fixed Sine vs. Bode 2

- what would you expect the steady-state input/output relationship to look like?



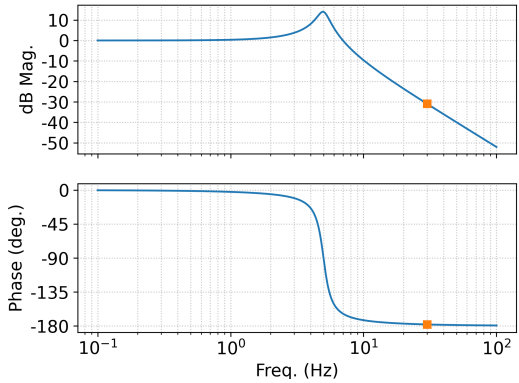
Fixed Sine vs. Bode 2 (answer)



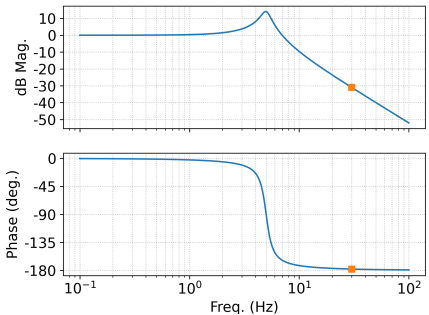


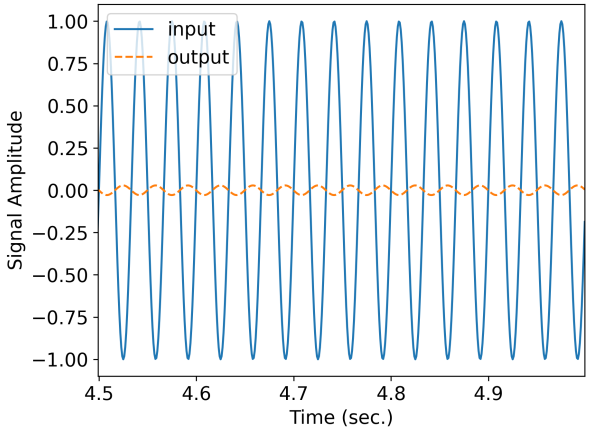
Fixed Sine vs. Bode 3

- what would you expect the steady-state input/output relationship to look like?



Fixed Sine vs. Bode 30 (answer)





Bode Sketching

My claim

Learning to sketch a Bode plot by hand will help you learn to read them.

- forces you to think about the details of the Bode plot
- focuses your thinking on the connections between the terms in the TF and their effects on the Bode plot

Our First Bode Sketching Example

Building block 1: a pure integrator

$$G_1(s) = \frac{2\pi}{s}$$

- multiplication my s is the same as differentiation, so dividing by s is the same as integrating

Bode Plot Table for $G_1(s)$

$$G_1(s) = \frac{2\pi}{s}$$

f	ω	s	$G(j\omega)$	M	dB	phase
0.01						
0.1						
1						
10						
100						
1000						
10000						

Deductive Reasoning

- Any polynomial with real coefficients can be factored into a product of first and second order terms.
- Therefore, the numerator and denominator of a transfer function can be factored into products of first and second order terms.

Dynamic systems terminology

- The roots of the denominator of a transfer function are called the poles.
- The roots of the numerator of a transfer function are called the zeros.

The six building blocks of transfer functions

- Based on the previous two slides, a transfer function is composed of some product of first and second-order poles along with first and second-order zeros
  - You might think that that leaves us with only four things but poles or zeros at the origin have special names and special significance
- So, the six building blocks are
  1. Pure integrator
  2. Pure differentiator
  3. First order pole
  4. First order zero
  5. Second order poles
  6. Second order zeros

Key Idea

- If we understand how each of the six building blocks affects a Bode plot, then we can sketch the Bode plot of any transfer function.
- If we can recognize the impact of each of the building blocks on a Bode plot, we can infer the transfer function from a Bode plot.

Building Block 2: Pure Diff.

$$G_2(s) = \frac{s}{1} = s$$

- what would happen if s increased by a factor of 10?
- what is the phase? does the phase change with frequency?

f	ω	s	$G(j\omega)$	M	dB	phase
0.01						
0.1						
1						
10						
100						
1000						
10000						

Building block 3: first-order pole

$$G_3(s) = \frac{p}{s+p}$$

- What happens to the magnitude and phase of $G_3(s)$ as the input frequency increases?
- two important frequency ranges:
  - low frequency ($\omega << p$)
  - high frequency ($\omega >> p$)

Bode Plot Table for $G_3(s)$

$$G_3(s) = \frac{p}{s+p}$$

assume $p = 10 \cdot 2 \cdot \pi$

f	ω	s	$G(j\omega)$	M	dB	phase
0.01						
0.1						
1						
10						
100						
1000						
10000						

Building Block 4: First Order Zero

$$G_4(s) = \frac{s+a}{a}$$

- two regions:
  - $\omega << a$
  - $\omega >> a$

Bode Plot Table for $G_4(s)$

$$G_4(s) = \frac{s+a}{a}$$

assume $a = 10 \cdot 2 \cdot \pi$

f	ω	s	$G(j\omega)$	M	dB	phase
0.01						
0.1						
1						
10						
100						
1000						
10000						

Building block 5: second-order (complex) poles

$$G_5(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- What happens to the magnitude and phase of $G_5(s)$ as the input frequency increases?
- three important frequencies ranges:
  - low frequency ($\omega << \omega_n$)
  - high frequency ($\omega >> \omega_n$)
  - near resonance ($\omega \approx \omega_n$)

Bode Plot Table for $G_5(s)$

$$G_5(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

assume $\omega_n = 10 \cdot 2 \cdot \pi$ and $\zeta = 0.1$

f	ω	s	$G(j\omega)$	M	dB	phase
0.01						
0.1						
1						
10						
100						
1000						
10000						

Building Block 6: Under-Damped Second-Order Zero

$$G_6(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{\omega_n^2}$$

- What happens to the magnitude and phase of $G_6(s)$ as the input frequency increases?
- three important frequencies ranges:
  - low frequency ($\omega << \omega_n$)
  - high frequency ($\omega >> \omega_n$)
  - near resonance ($\omega \approx \omega_n$)

Bode Plot Table for $G_6(s)$

$$G_6(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{\omega_n^2}$$

assume $\omega_n = 10 \cdot 2 \cdot \pi$ and $\zeta = 0.1$

f	ω	s	$G(j\omega)$	M	dB	phase
0.01						
0.1						
1						
10						
100						
1000						
10000						

Bode Sketching Algorithm

- find all poles and zeros
  - convert them to Hz
  - sort them from lowest to highest frequency
- identify the low frequency behavior
  - pure int, pure diff. or neither
    - first or second order int. or diff. possible
- work your way up in frequency
- when you reach a pole or zero, increment or decrement the slope **and** phase
- continue until you are above the highest frequency pole or zero
- double check the high frequency slope and phase
  - slope: $-20(n-m)$ dB/dec
  - phase: $-90^\circ(n-m)$